# Prem Kumar T

+91-9493437247 | premkumarthirupati@gmail.com | Hyderabad, India
premt.me | GitHub | LinkedIn | LeetCode

## Education

**CMR Institute of Technology** — Hyderabad, India
*B.Tech in Computer Science (AI & ML) – CGPA: 8.86/10* — *Aug 2023 – May 2027*

**Sri Chaitanya Junior College** — Hyderabad, India
*Intermediate (MPC – Science) – Score: 84.8%* — *Jun 2020 – May 2022*

## Technical Skills

**Languages:** C++, Python, JavaScript, Java, SQL
**AI, LLMs & ML:** Qwen 2.5 Coder, Gemini API, OpenRouter, Scikit-learn, Pandas, NumPy, AST Context Injection
**Backend & Systems:** Node.js, Express.js, Django, REST APIs, Redis (L1/L2 Caching), Rate Limiting
**Frontend & Databases:** React.js, Tailwind CSS, MongoDB, MySQL
**Core & Developer Tools:** Data Structures & Algorithms (DSA), Git, GitHub, Postman, Cursor

## AI Engineering & Full-Stack Projects

**SkillForge – AI-Powered Code Sandbox & Course Marketplace SaaS** — Jan 2026 – Apr 2026
*Full-Stack & AI Developer | Node.js, React.js, Qwen 2.5 32B, Gemini, Redis, MongoDB* — *Live Demo | GitHub*

- **Context-Aware AI Coding Mentor:** Integrated **Qwen 2.5 Coder 32B** and **Gemini** with dynamic AST analysis and sandbox test telemetry to deliver graduated hints, cutting learner debugging time by **45%** in **under 1s**.
- **Tiered L1/L2 Caching:** Architected an In-Memory L1 and Redis L2 caching pipeline with TTL expiry, async write-backs, and invalidation hooks, cutting MongoDB read load by **60%+** and API latency from **120ms to 15ms**.
- **Security, RBAC & Rate Limiting:** Secured **3 user roles** via dual-token JWTs (**HTTP-only refresh rotation**) and a 4-tier rate limiter (**200 req/15m** global cap) across Auth and Payment routes to prevent DDoS and bot abuse.
- **Creator Marketplace & Payments:** Built **15+ REST APIs** with compound MongoDB indexes for course publishing, a **350ms debounced** vanity profile engine, and Razorpay checkout with **SHA-256** cryptographic verification.

**Deepfake Detection System – Multi-Model Evaluation & Inference Pipeline** — Oct 2025 – Jan 2026
*Machine Learning Lead | Python, Scikit-learn, Pandas, NumPy, Django, MySQL* — *GitHub*

- **Multi-Model Benchmarking:** Designed a Python inference engine comparing and combining **3 ML estimators** (Naive Bayes, Linear SVC, Logistic Regression) via a **Voting Classifier** for high-precision media verification.
- **High-Volume Vectorization:** Vectorized **22,900+ dataset records** using Scikit-learn token frequency extraction, transforming unstructured text and metadata into high-dimensional sparse matrices for rapid inference.
- **Inference Logging & Telemetry:** Integrated a relational **MySQL** layer to persist real-time model predictions across **100% of user queries** and built evaluation dashboards tracking prediction distributions and deepfake ratios.

**Backend E-Commerce System – Scalable Retail API Engine** — Jun 2025 – Sep 2025
*Backend Developer | Node.js, Express.js, MongoDB, REST APIs* — *GitHub*

- **Modular Service Architecture:** Constructed **20+ REST APIs** across **3 decoupled layers** (controller, service, and data-access) to manage product catalogs, dynamic cart state, and end-to-end order lifecycle transitions.
- **Cryptographic Auth & Recovery:** Hardened account security using salted password hashing, route authorization middleware, and a **token-based password reset** workflow enforcing a strict **15-minute expiry window**.
- **Admin RBAC & Schema Validation:** Enforced role-based inventory controls and strict database schema validation rules across **4 core modules** to block malformed payloads and guarantee transactional consistency.

## Achievements & Certifications

- **Algorithms & Problem Solving (DSA):** Achieved a **1750+ contest rating** on LeetCode (**Top 9% globally**) across timed algorithmic contests by optimizing complex data structures for strict latency and memory bounds.
- **Smart Coder Certification** – Smart Interviews (Feb 2026) | View Certificate
- **React.js Course Completion Certificate** – Frontend Development (Aug 2025) | View Certificate